

Matrices

4.11 Diagonalization

If a $n \times n$ matrix A has a basis of eigenvectors then $D = X^{-1}AX$ is a diagonal matrix, with eigenvalues of A as the entries on the main diagonal. (principal diagonal) and X is the matrix with eigenvectors as column vectors.

$$D = X^{-1}AX$$

Question: Find the eigenvalues and diagonalized form of the matrix $A = \begin{bmatrix} 7.3 & 0.2 & -3.7 \\ -11.5 & 1 & 5.5 \\ 17.7 & 1.8 & -9.3 \end{bmatrix}$.

$$\text{Solution: } A - \lambda I = \begin{bmatrix} 7.3 - \lambda & 0.2 & -3.7 \\ -11.5 & 1 - \lambda & 5.5 \\ 17.7 & 1.8 & -9.3 - \lambda \end{bmatrix}$$

$$= (7.3 - \lambda) [\lambda^2 + 8.3\lambda - 19.2] - 1.92 - 2\lambda + 142.08 - 65.49\lambda$$

$$= \lambda^3 + \lambda^2 - 12\lambda = 0 \Rightarrow \lambda(\lambda^2 + \lambda - 12) = 0$$

$$\lambda_1 = -4, \lambda_2 = 3, \lambda_3 = 0$$

For eigenvector:

$$\lambda_1 = 3:$$

$$A - 3I = \begin{bmatrix} 4.3 & 0.2 & -3.7 \\ -11.5 & -2 & 5.5 \\ 17.7 & 1.8 & -12.3 \end{bmatrix}$$

$$(A - 3I)X = \begin{bmatrix} 4.3 & 0.2 & -3.7 \\ -11.5 & -2 & 5.5 \\ 17.7 & 1.8 & -12.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$4.3x_1 + 0.2x_2 - 3.7x_3 = 0 \quad (1)$$

$$-11.5x_1 - 2x_2 + 5.5x_3 = 0 \quad (2)$$

$$17.7x_1 + 1.8x_2 - 12.3x_3 = 0 \quad (3)$$

After solving the above equations we will get,

$$\Rightarrow x_1 = x_3$$

$$x_2 = -3x_1$$

$$\text{Let } x_1 = -1, x_2 = 3, x_3 = -1$$

$$\Rightarrow \lambda_1 = 3 \Rightarrow X_1 = \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}$$

For $\lambda_2 = -4$:

$$[A + 4I]X_2 = \begin{bmatrix} 11.3 & 0.2 & -3.7 \\ -11.5 & 5 & 5.5 \\ 17.7 & 1.8 & -5.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$11.3x_1 + 0.2x_2 - 3.7x_3 = 0$$

$$-11.5x_1 + 5x_2 + 5.5x_3 = 0$$

$$17.7x_1 + 1.8x_2 - 5.3x_3 = 0$$

After solving the above three equations we will get

$$x_3 = 3x_1 \quad \& \quad x_1 = -x_2$$

$$\text{Let } x_1 = 1, \Rightarrow X_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

Now for $\lambda_3 = 0$

$$[A - 0]X_3 = \begin{bmatrix} 7.3 & 0.2 & -3.7 \\ -11.5 & 1 & 5.5 \\ 17.7 & 1.8 & -9.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$7.3x_1 + 0.2x_2 - 3.7x_3 = 0$$

$$-11.5x_1 + x_2 + 5.5x_3 = 0$$

$$17.7x_1 + 1.8x_2 - 9.3x_3 = 0$$

By solving the above equations:

$$x_1 = 2x_2, \quad x_3 = 4x_2$$

$$\text{Let } x_2 = 1 \Rightarrow \text{Then } x_1 = 2 \Rightarrow x_3 = 4$$

$$\therefore X_3 = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

$$X = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix} \Rightarrow |X| = 7 - 1(13) + 2(8) = 10$$

$$X^{-1} = \frac{\text{adj}X}{|X|}$$

$$\text{adj}X = \begin{bmatrix} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} & \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} \\ -\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ -1 & 4 \end{vmatrix} & -\begin{vmatrix} -1 & 1 \\ -1 & 3 \end{vmatrix} \\ \begin{vmatrix} -1 & 2 \\ -1 & 1 \end{vmatrix} & -\begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 1 \\ 3 & -1 \end{vmatrix} \end{bmatrix}$$

$$\text{adj}X = \begin{bmatrix} -7 & -13 & 8 \\ 2 & -2 & 2 \\ 1 & 7 & -2 \end{bmatrix}^T = \begin{bmatrix} -7 & 2 & 1 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

$$X^{-1} = \frac{\text{adj}X}{|X|} = \begin{bmatrix} -0.7 & 0.2 & 0.1 \\ -1.3 & -0.2 & 0.7 \\ 0.8 & 0.2 & -0.2 \end{bmatrix}$$

$$D = X^{-1}AX = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Important property:

$$D = X^{-1}AX$$

- $A = XDX^{-1}$
- $A^n = XD^nX^{-1}$

Example: $A^{100} = XD^{100}X^{-1}$

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \Rightarrow D^{100} = \begin{bmatrix} \lambda_1^{100} & 0 & 0 \\ 0 & \lambda_2^{100} & 0 \\ 0 & 0 & \lambda_3^{100} \end{bmatrix}$$

$$\Rightarrow D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \Rightarrow D^{100} = \begin{bmatrix} \lambda_1^{100} & 0 \\ 0 & \lambda_2^{100} \end{bmatrix}$$

$$e^D = \begin{bmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{bmatrix} \Rightarrow \cos D = \begin{bmatrix} \cos \lambda_1 & 0 \\ 0 & \cos \lambda_2 \end{bmatrix}$$

- The inner product is always normalized.